

AirSense

AQI Prediction & Analysis Platform

Real-time Air Quality Monitoring, Forecasting & Smart Health Recommendations



Overview

AirSense is a comprehensive Air Quality Index (AQI) prediction and analysis platform that provides real-time monitoring, forecasting, smart health recommendations, interactive maps, and community reporting. The platform leverages Gemini AI for intelligent analysis and predictions.

Key Features:

-  **Real-time AQI Monitoring** - Track air quality across multiple cities
-  **Interactive Dashboard** - Visualize air quality data with charts and maps
-  **Interactive Maps** - Leaflet-based geographic visualization
-  **AQI Forecasting** - Predict future air quality trends
-  **Health Recommendations** - AI-powered health advisories based on AQI
-  **Community Feed** - Share and view community reports
-  **Simulator** - Test scenarios and predictions
-  **Carbon Calculator** - Track carbon emissions and footprint
-  **City Comparison** - Compare air quality across cities
-  **Source Analysis** - Identify pollution sources

-  **Alerts Manager** - Get notified of dangerous AQI levels
-  **Data Provenance** - Track data sources and reliability
-  **Admin Console** - System administration and monitoring
-  **Awareness Hub** - Educational content about air quality



Tech Stack

- **Frontend:** React 19, TypeScript, Vite
- **Styling:** Tailwind CSS 4, Motion animations
- **Maps:** Leaflet
- **Charts:** Recharts
- **AI:** Google Gemini API
- **Icons:** Lucide React
- **Validation:** Dotenv for environment management
- **Build Tool:** Vite with Tailwind plugin



Project Structure

```
airsense-3/
├── src/
│   ├── components/           # React components
│   │   ├── Dashboard.tsx     # Main dashboard view
│   │   ├── InteractiveMap.tsx
│   │   ├── CityDetail.tsx
│   │   ├── CityComparison.tsx
│   │   ├── Simulator.tsx
│   │   └── SourceAnalysis.tsx
```

```
| | └─ CarbonCalculator.tsx
| | └─ CommunityFeed.tsx
| | └─ AwarenessHub.tsx
| | └─ AlertsManager.tsx
| | └─ DataProvenance.tsx
| | └─ AdminConsole.tsx
| | └─ Sidebar.tsx
| | └─ Header.tsx
| └─ context/
| | └─ AppContext.tsx # Global app state management
| └─ data/
| | └─ mockData.ts # Mock data for testing
| └─ types/
| | └─ index.ts # TypeScript type definitions
| └─ utils/
| | └─ aqiCalculations.ts # AQI calculation utilities
| └─ App.tsx # Root component
| └─ main.tsx # Entry point
| └─ index.css # Global styles
└─ package.json
└─ tsconfig.json
└─ vite.config.ts
└─ index.html
└─ metadata.json
```

Getting Started

Prerequisites

- Node.js 16+
- npm or yarn
- Gemini API Key

Installation

1. **Clone or download the project:**

```
cd aairsense-3
```

2. Install dependencies:

```
npm install
```

3. Configure environment variables:

Create or update `.env.local` file in the root directory:

```
VITE_GEMINI_API_KEY=your_gemini_api_key_here
```

4. Start development server:

```
npm run dev
```

The app will be available at <http://localhost:3000>

Build for Production

```
npm run build
```

Output will be in the `dist/` directory.

Preview Production Build

```
npm run preview
```



Key Components

Component	Purpose
Dashboard	Main overview of air quality data
InteractiveMap	Leaflet-based geographic visualization
CityDetail	Detailed AQI info for a specific city
CityComparison	Compare AQI across multiple cities
Simulator	Scenario testing and predictions
SourceAnalysis	Identify and analyze pollution sources
CarbonCalculator	Calculate and track carbon footprint
CommunityFeed	User-generated air quality reports
AwarenessHub	Educational content about air quality
AlertsManager	Configure and manage AQI alerts
DataProvenance	Track data sources and accuracy
AdminConsole	System administration tools



AQI Bands

The platform uses standard AQI classifications:

Band	AQI Range	Description
Good	0-50	Air quality is satisfactory
Moderate	51-100	Acceptable; some pollutants may be a concern
Sensitive	101-150	Members of sensitive groups may experience health effects
Unhealthy	151-200	General public may begin to experience health effects
Very Unhealthy	201-300	Health alert; everyone may begin to experience serious effects
Hazardous	301+	Health warning; emergency conditions



Data Tracked

The platform monitors pollutants including:

- PM_{2.5} (Fine particulate matter)
- PM₁₀ (Coarse particulate matter)
- NO₂ (Nitrogen dioxide)
- SO₂ (Sulfur dioxide)
- CO (Carbon monoxide)
- O₃ (Ozone)

Plus weather data:

- Temperature, humidity, wind speed
- Pressure, UV index, visibility

- Weather conditions

Development Scripts

```
npm run dev      # Start dev server on port 3000
npm run build    # Build for production
npm run preview  # Preview production build
npm run lint     # Run TypeScript type checking
npm run clean    # Clean dist and server artifacts
```

Styling

The project uses **Tailwind CSS 4** with dark mode support. Global styles are in `src/index.css`.

Color scheme:

- Light mode: Clean white backgrounds with dark text
- Dark mode: Dark (#191919) backgrounds with light text

AI Integration

AirSense integrates with **Google Gemini API** for:

- Health recommendations based on AQI levels
- Air quality predictions and forecasting
- Pollution source analysis
- Community insights and awareness content

Ensure your `VITE_GEMINI_API_KEY` is properly configured.



Type Safety

All components use TypeScript with strict type checking enabled. Key types are defined in `src/types/index.ts`.



Permissions

The application requests:

- **Geolocation** - To provide location-based air quality data



Configuration Files

- `vite.config.ts` - Vite build configuration
- `tsconfig.json` - TypeScript configuration
- `tailwind.config.ts` - Tailwind CSS configuration (if exists)
- `.env.local` - Environment variables (not committed to repo)



License

This project is licensed under the Apache License 2.0. See LICENSE file for details.



Contributing

Contributions are welcome! Please follow these steps:

1. Create a feature branch
2. Make your changes
3. Test thoroughly

4. Submit a pull request

Support

For issues, feature requests, or questions, please create an issue in the project repository.

Last Updated: 2026-09-01

Version: 0.0.0

Platform: React + Vite + Tailwind CSS